

Water Tanks Monitor System — Technical Guide & Support Manual

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1. About this guide

This guide covers installing, configuring, operating and troubleshooting the Water Tanks Monitor System (WTMS), firmware **2.10.0**, by woodyouloveit.com. It is written for installers and site administrators; residents who only view the dashboard need section 7.1 and section 10.

What the system does. A waterproof ultrasonic sensor mounted above the water measures the distance to the surface. An ESP32 or ESP8266 turns that distance into a fill level, shows it on a WS2812B LED strip (red at the bottom, fading to green at the top), and serves a web dashboard over Wi-Fi with live level, litres, analytics, a PDF report and a connectivity log.

Main parts

- **Controller:** ESP32 DevKit V1 (30-pin) or ESP8266 NodeMCU v1.0 (30-pin). One firmware builds for both.
- **Sensor:** AJ-SR04M (board marked SR04M-2) or JSN-SR04T, with a waterproof probe on a cable.
- **Display:** WS2812B LED strip, 1 to 300 LEDs.
- **Dashboard:** built into the firmware, works offline on the device's own hotspot or on the home Wi-Fi. Tabs: Overview, Analytics, Setup (admin), Log (admin).
- **Storage:** settings in EEPROM (flash); level history, fills and the log in LittleFS (flash), kept across restarts.

Firmware files (all in the `water_level_indicator` sketch folder):

`water_level_indicator.ino`, `dashboard_html.h`, `event_log.h`, `history.h`, `auth.h`.

2. Technical specifications

Figures are for firmware 2.10.0. Sensor accuracy is the typical value for the AJ-SR04M / JSN-SR04T class; your installation may differ.

2.1 Controllers

Item	ESP32 DevKit V1 (30-pin)	ESP8266 NodeMCU v1.0 (30-pin)
Module	ESP-WROOM-32	ESP-12E
Flash	4 MB	4 MB
CPU clock	80 MHz (power saving) or 240 MHz (performance)	80 MHz
Logic level	3.3 V (not 5 V tolerant)	3.3 V (not 5 V tolerant)
Supply	5 V on VIN	5 V on VIN
Program space	1.2 MB or 1.9 MB per slot, set by the partition scheme	about 1 MB area (OTA needs a program under about 500 KB)
Built-in button used	BOOT (GPIO0)	FLASH (GPIO0)

2.2 Sensor (AJ-SR04M / JSN-SR04T)

Item	Value
Supply	3.0 to 5.5 V (installed on 3.3 V from the board's 3V3 pin)
Operating mode	Trigger / echo (mode 1, resistor position R19 empty)
Trigger pulse	100 μ s default; 50 to 500 μ s works; 10 to 20 μ s is ignored by the AJ-SR04M
Blind zone	about 20 to 25 cm from the probe face
Range used by the firmware	up to 450 cm; echoes longer than 27 ms (about 460 cm) are treated as no echo
Resolution	about 1 cm from the sensor; shown to 0.1 cm after the median filter
Distance formula	$\text{distance (cm)} = \text{echo time } (\mu\text{s}) \times 0.0343 \div 2$

2.3 LED strip and power

Item	Value
Strip type	WS2812B, 5 V
Number of LEDs	1 to 300 (set in the dashboard)
Data line	3.3 V from the board through a 330 Ω resistor
Current	up to 60 mA per LED at full white; the level display (red/green only) uses less; the dashboard shows an estimate
Power supply	5 V, at least 2 A for 30 LEDs or 3 A for 60 LEDs
Supply capacitor	1000 μ F, 10 V or higher, across the strip's 5 V and GND

2.4 Wi-Fi and network

Item	Value
Radio	2.4 GHz, 802.11 b/g/n
Transmit power	Low 8.5 dBm, Medium 13 dBm (default), High 19.5 dBm (20.5 dBm on ESP8266)
Hotspot name	WTMS-XXXX (chip ID), or WTMS <building> <tank> once named; max 32 characters
Hotspot security	open network; the dashboard itself needs the admin password for any change
Hotspot address	http://192.168.4.1 (captive portal opens the dashboard)
Home network address	http://tank-<building>-<tank>.local or the IP address
Web server	HTTP on port 80; dashboard page about 125 KB, served from flash

2.5 Storage and measurement

Item	Value
Settings storage	EEPROM area, 512 bytes (kept across restarts and OTA updates)
Level history	1 record every 2 min, 14 days (2 files × 5,040 records × 8 bytes)
Fill records	2 files × 100 records
Connectivity log	2 files × 12 KB, about 300 to 400 events
Total flash data	about 107 KB at most
Sensor reading	every 5 s (Eco, default), 2 s, 0.5 s or custom 0.2 to 30 s
Level filter	median of 5 readings (2 s and slower) or 7
Fill detection	from 10-second samples

3. Bill of materials

One set per tank.

Qty	Part	Specification	Notes
1	Controller	ESP32 DevKit V1 (30-pin) or NodeMCU ESP8266 (30-pin)	ESP32 recommended: OTA has more room
1	Ultrasonic sensor	AJ-SR04M (SR04M-2) or JSN-SR04T, with probe and cable	Check that position R19 on the sensor board is empty
1	LED strip	WS2812B, 5 V, 30 or 60 LEDs (1 to 300 supported)	
1	Power supply	5 V DC, 2 A (30 LEDs) or 3 A (60 LEDs)	Do not power the strip through the board or USB
1	Resistor	330 Ω	In the LED data line, close to the strip

Qty	Part	Specification	Notes
1	Capacitor	1000 μ F electrolytic, 10 V or higher	Across the strip's 5 V and GND; stripe to GND
2	Female headers	15-pin	Makes the board removable
3	Screw terminals	4-pin (sensor), 3-pin (strip), 2-pin (power)	
1	Perfboard	about 7 × 9 cm	
1	Stand pipe (loft or shallow tanks)	PVC, at least 25 cm tall and 75 mm inside diameter	Keeps the full water line out of the blind zone
1	Enclosure	ventilated, away from direct sun	
—	Hook-up wire	20 to 22 AWG for 5 V and GND, thinner for signals	

The 1 k Ω and 2 k Ω echo divider from earlier designs is not needed: the sensor runs on 3.3 V, so its echo output is already 3.3 V.

4. Wiring

The sensor runs on 3.3 V from the board, so its echo wire connects directly. The LED strip and the board's VIN take 5 V from the external supply, and all grounds are joined.

4.1 ESP32 DevKit V1

The sensor uses four neighbouring pins (3V3, GND, D15, D2), so one 4-pin connector does the job.

From	ESP32 pin	Notes
Sensor "5V" pin	3V3	3.3 V only, never the 5 V supply
Sensor GND	GND (next to 3V3)	
Sensor RX (TRIG)	D15	Must be D15
Sensor TX (ECHO)	D2	Must be D2

From	ESP32 pin	Notes
LED strip DIN	G16 (RX2)	Through 330 Ω
5 V supply +	VIN	Also to the strip's 5 V
5 V supply –	GND	Also to the strip's GND

Keep D15 = TRIG and D2 = ECHO. Both are boot strapping pins. D2 must be LOW while uploading over USB; the sensor's echo output rests LOW, so uploads work. Swapped, the sensor's RX input can hold D2 HIGH and uploads fail with "wrong boot mode". A blue LED on D2, if the board has one, flickers with each reading; that is normal.

4.2 ESP8266 NodeMCU

From	NodeMCU pin	Notes
Sensor "5V" pin	3V3	3.3 V only when ECHO is wired directly
Sensor GND	GND	
Sensor RX (TRIG)	D5 (GPIO14)	
Sensor TX (ECHO)	D6 (GPIO12)	Direct; with the sensor on 5 V, add a 1 k Ω / 2 k Ω divider
LED strip DIN	D2 (GPIO4)	Through 330 Ω
5 V supply +	VIN	Also to the strip's 5 V
5 V supply –	GND	Also to the strip's GND

Do not use D3, D4 or D8 for external wiring: the board reads them at boot.

4.3 Wiring rules

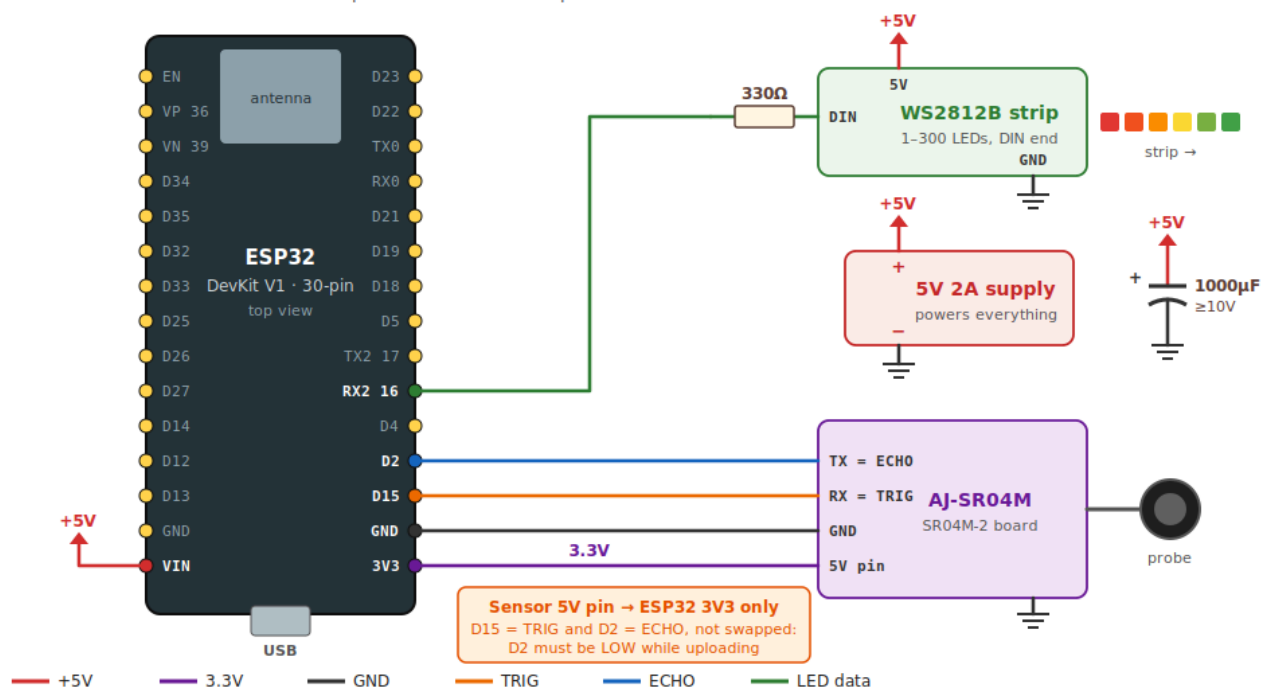
1. Power the strip from the 5 V supply, never through the board or its USB port.
2. Fit the 1000 μ F capacitor across the strip's 5 V and GND, stripe to GND.
3. Put the 330 Ω resistor in the data line, close to the strip.
4. Join all grounds: supply, board, strip and sensor.
5. For strips longer than about 60 LEDs, also feed 5 V and GND to the far end.

6. The sensor's pins are labelled RX and TX on the AJ-SR04M board: RX is the trigger input, TX is the echo output.

4.4 Wiring diagrams

Water level indicator: ESP32 DevKit V1 (30-pin)

Sensor on 3V3 · GND · D15 · D2 (one 4-pin connector) · LED strip G16

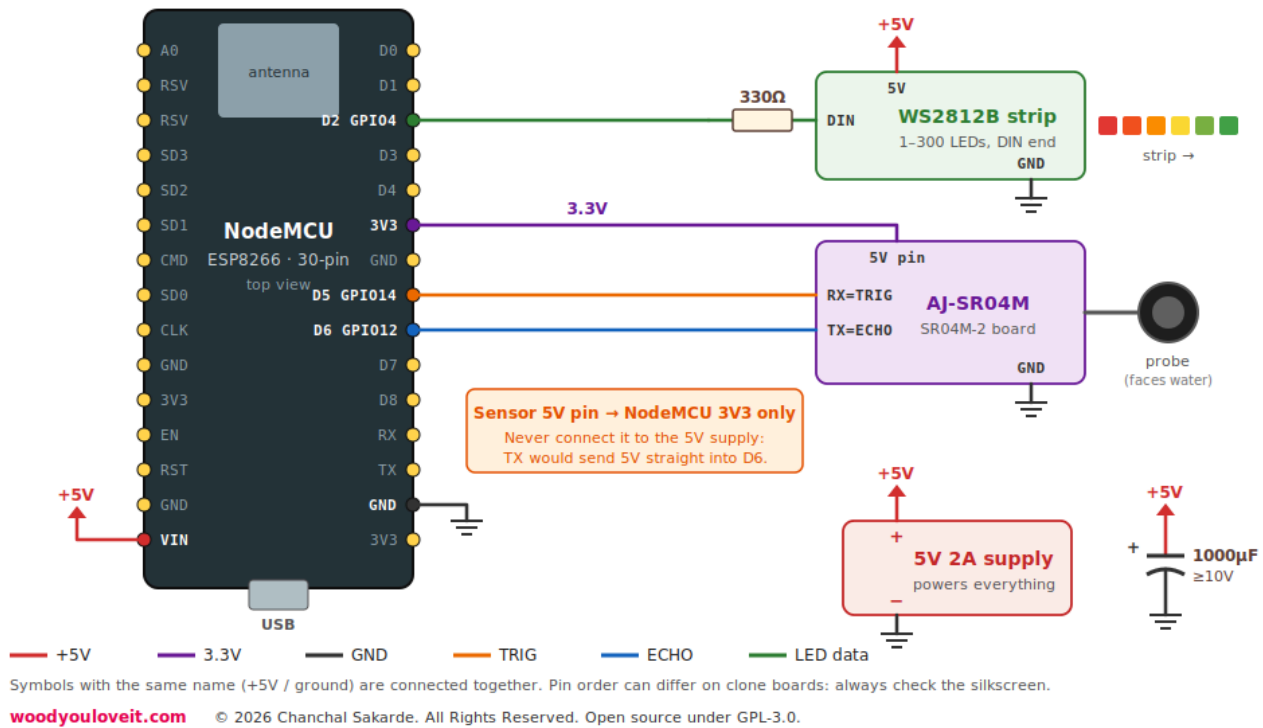


Symbols with the same name (+5V / ground) are connected together. Pin order can differ on clone boards: always check the silkscreen.

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NodeMCU ESP8266 + AJ-SR04M on 3.3V (no divider)

LED D2 (GPIO4) · sensor RX = TRIG from D5 (GPIO14) · sensor TX = ECHO to D6 (GPIO12)



5. Mechanical installation

Mount the probe at the top of the tank, pointing straight down at the water, with the full water line at least 25 cm below the probe face.

5.1 Probe position

- Centre of the lid or manhole, pointing vertically down.
- Clear of inlet pipes, overflow pipes, ladders, floats and tank walls, which reflect sound and give false readings.
- Away from the inlet stream, where splashing and foam scatter the echo.
- The cable from the probe to the sensor board must not be cut or extended with thin wire; route it away from the LED strip's power wires.

5.2 Blind zone and stand pipe

The sensor cannot measure anything closer than about 20 to 25 cm. If the full water line is closer than that, the level near full is wrong or missing.

Tank	Typical height	Mounting
Overhead (roof) tank, 500 L and larger	80 cm or more	On the lid, if the full line is 25 cm or more below it
Loft / bathroom tank, 150 to 1000 L	385 to 685 mm	On a stand pipe above the manhole
Underground sump	varies	On the manhole cover, probe away from the inlet and walls

Stand pipe for shallow tanks: a PVC pipe at least 25 cm tall and 75 mm inside diameter, standing on the manhole, with the probe at the top pointing down through it. A narrower pipe causes echoes from the pipe wall.

Resolution on shallow tanks: 1 cm of water is about 2.5 % of a loft tank (about 6 L in a 225 L tank). The sensor resolves about 1 cm, so expect the level to move in steps of that size.

5.3 Controller and enclosure

- Keep the controller and sensor board dry and out of direct sun.
- Give the enclosure a few ventilation holes; the board and its 3.3 V regulator run warm.
- Mount the LED strip where it can be seen, running bottom to top (set "Strip mounted top-down" in the dashboard if the first LED is at the top).
- Leave the board's USB port reachable for the first upload, and the BOOT button reachable for the on-demand hotspot and password reset.

6. Software and flashing

The firmware is built with the Arduino IDE and needs one library, Adafruit NeoPixel; Wi-Fi, web server, DNS, mDNS, LittleFS and OTA come with the board package.

6.1 Preparing the Arduino IDE

1. Install the Arduino IDE (2.x).
2. Add the board package:
 - ESP32: Boards Manager → "esp32" by Espressif.

- ESP8266: File → Preferences → Additional Boards Manager URLs → `http://arduino.esp8266.com/stable/package_esp8266com_index.json`, then Boards Manager → "esp8266".

3. Library Manager → install **Adafruit NeoPixel**.

4. Open `water_level_indicator/water_level_indicator.ino`. The other four `.h` files open as tabs and must stay in the same folder.

6.2 Tools settings, ESP32 Dev Module

Setting	Use	Why
Board	ESP32 Dev Module	
Partition Scheme	Default 4MB with spiiffs (1.2MB APP / 1.5MB SPIFFS) if the sketch fits; otherwise Minimal SPIFFS (1.9MB APP with OTA / 190KB SPIFFS)	Both keep OTA and a storage area for the log and history
Erase All Flash Before Sketch Upload	Disabled	Enabled wipes settings, Wi-Fi, calibration, history and logs on every upload
CPU Frequency	240 MHz or 80 MHz	Either; power saving switches to 80 MHz by itself
Core Debug Level	None	Keeps Serial Monitor readable
Upload Speed	921600	Use 115200 if uploads fail
Flash Size	4MB (32Mb)	

Partition schemes to avoid: Huge APP and all No OTA schemes (no OTA), Minimal 1.3MB (no OTA), any FATFS scheme (no "spiiffs" area, so the log and history are kept in memory only), 8M and 16M schemes (the board has 4 MB), RainMaker.

Choosing between the two good schemes: press Verify and read `Sketch uses N bytes ... Maximum is ... bytes`. Under about 1,250,000 bytes → Default 4MB with spiiffs. Larger → Minimal SPIFFS; its 190 KB is enough, as the firmware stores at most about 107 KB.

6.3 Tools settings, NodeMCU 1.0 (ESP-12E Module)

Setting	Use
Board	NodeMCU 1.0 (ESP-12E Module)
Flash Size	4MB (FS:2MB OTA:~1019KB)
Erase Flash	Only Sketch
Upload Speed	115200 or 921600

6.4 First upload

1. Connect the board by USB, select its Port.
2. Press Upload. If an ESP32 upload stalls at "Connecting...", hold BOOT until writing starts; if it fails with "wrong boot mode", unplug the sensor connector, upload, and plug it back.
3. Open Serial Monitor at 115200 baud. The first lines show `Water Tanks Monitor System 2.10.0 | woodyouloveit.com | board esp32` and the copyright line.
4. On a first start, or after a flash erase, the log shows "Flash storage was blank and has been formatted" and Boot #1. That is normal.

Changing the partition scheme needs a USB upload (OTA cannot change it). Settings are kept; history and log start fresh.

7. First-time setup

Setup takes about 15 minutes: connect to the device's hotspot, log in as admin, then work down the Setup tab and finish by exporting a settings backup.

7.1 Opening the dashboard

Where you are	Address
Connected to the device's hotspot (WTMS-XXXX or WTMS <building> <tank>)	opens by itself; otherwise http://192.168.4.1
On the same home Wi-Fi as the device	http://tank-<building>-<tank>.local, or the IP shown in Setup → Wi-Fi settings

The top LED of the strip blinks blue while the hotspot is on. Type the address with `http://`; `https://` pages and other websites cannot be redirected to the dashboard. On a phone, turn off mobile data while on the hotspot. Without logging in, anyone sees the Overview and Analytics tabs.

7.2 Admin login

1. Press **Admin** in the header.
2. Enter the default password `admin`.
3. Choose a new password (at least 6 characters, not "admin"). The dashboard does not continue until it is changed.

The Setup and Log tabs appear once logged in.

7.3 Step by step in the Setup tab

1. **Wi-Fi settings:** press Scan, pick the home network, enter its password, press Connect. The card shows the new address; in automatic hotspot mode the hotspot turns off 30 s later. Reconnect your phone to the home Wi-Fi and open that address.
2. **Site details:** society or organisation (for example NYATI), building (for example YUCCA) and tank name (for example Overhead Tank 1). Save, then **Restart device** to apply the new hotspot name and web address.
3. **Usage & tank:** Domestic or Commercial; location (overhead, loft / bathroom, underground sump, other); shape; a loft preset or custom dimensions; **water depth when full** (tank bottom to the overflow pipe, measured); rated capacity in litres. Save tank, then press **Recommended settings** and confirm.
4. **Calibration:** use one method from section 9.4: the current-level slider (no need to empty or fill the tank), measurements, or the Empty / Full buttons. While the Setup tab is open, the sensor is read every 0.5 s.

5. **Display & sensor:** number of LEDs on the strip, brightness (the card shows the supply current needed), low water alarm, strip direction, units for everyone (mm, cm or inch).
6. **Power & radio:** power saving on, hotspot **On demand**, transmit power Low or Medium.
7. **Update rate:** Eco (sensor every 5 s, dashboard every 10 s).
8. **Admin & backup:** press **Export settings** and keep the file.

7.4 Checking the installation

- Overview shows a plausible level and litres; "Tank at a glance" shows no warnings.
- Setup → Device shows Log storage: Flash (kept after restart).
- Pour or draw some water and watch the level follow within about 15 s (Eco).
- The next day, Analytics shows the level history and any fills.

8. Configuration reference

All settings are changed by an admin in the dashboard and stored in flash. Distances are stored in cm and tank sizes in mm, whatever unit is displayed.

Card	Setting	Default	Range / options
Site details	Society / organisation	empty	up to 48 characters
Site details	Building, Tank	empty	up to 32 characters each
Usage & tank	Usage	Domestic	Domestic, Commercial
Usage & tank	Tank location	Overhead	Overhead, Loft / bathroom, Underground sump, Other
Usage & tank	Tank shape	Rectangular	Rectangular, Vertical cylinder
Usage & tank	Tank size	Custom	Loft presets 150, 225, 270, 400, 500, 1000 L, or length × width × height / diameter × height, up to 20,000 mm
Usage & tank	Water depth when full	not set	up to 20,000 mm, not more than the tank height

Card	Setting	Default	Range / options
Usage & tank, Analytics	Capacity	0 (amounts in %)	0 to 1,000,000 L
Calibration	Empty distance	120 cm	up to 450 cm
Calibration	Full distance	25 cm	at least 15 cm; empty – full at least 10 cm
Display & sensor	Units for everyone	cm	mm, cm, inch (each browser can override)
Display & sensor	Number of LEDs	30	1 to 300
Display & sensor	LED brightness	150	5 to 255
Display & sensor	Low water alarm	10 %	0 to 50 %
Display & sensor	Strip mounted top-down	off	on / off
Display & sensor	Whole bar one colour	off	on / off
Display & sensor	Sensor trigger pulse	100 μ s	10 to 1000 μ s (AJ-SR04M: 50 to 500)
Analytics settings	Night check window	01:00 to 05:00	any start and end hour, not equal
Analytics settings	Leak limit	1.5 cm	drop in the night window
Analytics settings	Fill detection threshold	0.5 cm/min	minimum rise speed
Power & radio	Power saving	on	on / off (performance)
Power & radio	Hotspot	Automatic	Automatic, On demand (BOOT), Always on
Power & radio	Transmit power	Medium 13 dBm	Low 8.5, Medium 13, High 19.5 dBm

Card	Setting	Default	Range / options
Update rate	Profile	Eco	Eco, Balanced, Responsive, Custom
Update rate	Custom sensor interval	5 s	0.2 to 30 s
Update rate	Custom dashboard refresh	10 s	2 to 60 s
Admin	Password	admin (must be changed)	at least 6 characters

Recommended settings (Usage & tank card, applied on confirmation):

	Domestic	Commercial
Night leak check	01:00 to 05:00	22:00 to 06:00
Low water alarm	15 % (30 % for a sump)	25 % (30 % for a sump)
Leak limit	2 % of the water depth, at least 1 cm	same

The Defaults button (Display & sensor) resets calibration, display and analytics settings. It keeps the LED count, capacity, units, update rate, power, transmit power, hotspot mode, site, tank profile and Wi-Fi.

9. How measurement works

The level comes from the time an ultrasonic echo takes to return from the water surface, filtered by a median of several readings and scaled between the calibrated empty and full distances.

9.1 One reading

1. The controller holds TRIG (sensor RX) high for the trigger pulse, 100 μ s by default.
2. The sensor sends a burst of 40 kHz ultrasound and raises ECHO (sensor TX).
3. ECHO stays high until the echo returns; the controller times it.
4. Distance (cm) = echo time (μ s) $\times$ 0.0343 $\div$ 2. Example: water 50 cm away gives about 2.9 ms.

- No echo within 27 ms (about 460 cm), or a result over 450 cm, counts as a missed reading.

The AJ-SR04M ignores trigger pulses of 10 to 20 μ s; 50 to 500 μ s work. Change the pulse only for a different sensor or if "no echo" returns.

9.2 Timing

Profile	Sensor reading	Median of	Level follows a change within	"No echo" shown after	Open dashboard refresh
Eco (default)	every 5 s	5	about 15 s	about 15 s	every 10 s
Balanced	every 2 s	5	about 6 s	about 14 s	every 5 s
Responsive	every 0.5 s	7	about 2 s	about 5 s	every 2 s
Custom	0.2 to 30 s	5 at 2 s and slower, else 7	shown in the card	shown in the card	2 to 60 s

- While an admin has the Setup tab open, the sensor is read every 0.5 s (live mode) and returns to the profile 20 s after leaving.
- Independent of the profile: fill detection samples every 10 s, history is saved every 2 min, a health line is logged every 5 min.
- The dashboard only refreshes while a page is open; with no page open the device serves nothing.

9.3 From distance to level

- Level % = (empty distance – distance) ÷ (empty distance – full distance) × 100, limited to 0 to 100 %.
- Litres = level × rated capacity (assumes straight tank walls).
- The LED bar follows a smoothed level, so it does not flicker; the top lit LED is dimmed in proportion for a smooth bar.
- "No echo" (sensor fault) is declared after about 15 s of consecutive missed readings (3 to 10 readings).

9.4 Calibration methods

All three set the empty and full distances; the water depth D = empty – full.

Method	You provide	Calculation
Current-level slider	water depth D, how full the tank is now (p %)	empty = live reading + $p \times D$; full = empty - D
Measurements	water depth D, sensor face to full water line G	full = G; empty = G + D
Empty / Full buttons	the tank actually empty or full	the live reading is saved as that point

Every method previews the result and refuses a full line closer than 15 cm (warning below 22 cm) or an empty point beyond 450 cm.

9.5 Fill (motor) detection

- A fill starts when the level rises faster than the threshold (default 0.5 cm/min) by at least 1.5 cm.
- It ends when the rise slows or the tank is full.
- Fills shorter than 3 minutes or smaller than 3 cm are ignored.
- Every fill is counted as the motor; water from another source that fills the tank is counted too.

9.6 Water used

Every drop in level outside a fill, with a small dead band so sensor noise is not counted as use.

9.7 Night leak check

For each night, the level change in the night window (default 01:00 to 05:00) is compared with the leak limit (default 1.5 cm), when enough readings exist and no fill ran.

Result	Rule
OK	drop under half the limit
Watch	drop between half the limit and the limit
Water used (one drop)	drop over the limit, 70 % or more of it within one hour (a tap or flush)

Result	Rule
Possible leak	drop over the limit, slow and steady

10. Analytics and PDF reports

The Analytics tab and the PDF report are open to everyone; only saving analytics settings and clearing history need an admin. The charts are calculated in the browser from the device's stored history, in the viewer's local time.

10.1 Analytics tab

Part	Shows
Summary tiles	level now or filling (speed, time to full), last fill, empty → full time, motor time today, water used today, busiest hour, last night's check
Level history	24 h, 3, 7 or 14 days, fills and night windows shaded, values on hover or tap
Motor run time per day	last 7 days, plus the most recent fills
Water used per day	last 7 days
Usage by hour of day	average use per hour, busiest 3 hours highlighted
Night leak check	last 7 nights with result
Analytics settings (admin)	capacity, fill threshold, night window, leak limit, clear history

History needs the real time: it comes from the internet on home Wi-Fi, or from the browser when the dashboard is opened on the hotspot. The Overview tab also warns when last night's check found a possible leak.

10.2 PDF report

Press **Download PDF report** in the Analytics tab. The report covers the last 7 days in A4 and is made in the browser, so it also works offline.

Page	Content
1	logo, society, building and tank; details table; summary tiles; 7-day level chart; motor time and water used per day
2	usage by hour; night leak check; recent fills; how the numbers are calculated
3	Settings used for this report: calibration, capacity, resolution, units, alarm and leak limits, night window, fill threshold, sensor rate, power mode, firmware

Every page carries the copyright line, woodyouloveit.com, the page number, the firmware version and the Settings ID. The file is named `wtms-report-<building>-<tank>-<date>.pdf` . On the hotspot's pop-up window, open `http://192.168.4.1` in the normal browser first, as some pop-ups block downloads.

10.3 Settings ID

The Settings ID is an 8-character fingerprint of the configuration (for example `5B8AF761`). It appears in the Admin & backup card, in exported settings files and in the report. Matching the ID in a report with an exported settings file shows exactly which configuration produced the report's figures.

11. Access control and security

Anyone on the network can view the level and analytics; every change needs the admin password, and the device itself rejects admin actions from viewers (not only the dashboard).

	Viewer (no login)	Admin
Overview tab	yes	yes
Analytics tab, PDF report	yes	yes
Save analytics settings, clear history	view only	yes

	Viewer (no login)	Admin
Setup tab (all settings, calibration, Wi-Fi, backup, firmware update)	hidden	yes
Log tab	hidden	yes
Restart device, reset to defaults	no	yes
Units menu in the Water level card (own browser only)	yes	yes

11.1 Password

- Default `admin` ; it must be changed at the first login (at least 6 characters, not "admin").
- Change it later in Setup → Admin & backup → Change password.
- **Forgotten:** hold BOOT (ESP32) or FLASH (NodeMCU) for 10 seconds until the strip flashes red three times. The password returns to `admin` and all admin sessions end.
- After 5 wrong passwords, login is blocked for 1 minute. Logins, failures and password changes appear in the log.

11.2 How the login is protected

- The password never crosses the network. The browser receives a salt and a one-time challenge and returns a SHA-256 proof.
- The device stores only a salted SHA-256 hash of the password.
- A session ends after 12 hours without activity or when the device restarts; up to 3 sessions can be open.

11.3 Limits to be aware of

- The dashboard uses plain HTTP. On an untrusted network, someone watching the traffic could see the session token and the data. Keep the device on your own Wi-Fi.
- The hotspot is an open network. Viewers who join it can see the level; changes still need the password. Use the On demand hotspot mode so it is off most of the time.
- Exported settings files contain no passwords, but they include the home Wi-Fi name and the site names.

12. Power, heat and production settings

For a device that runs for years, use power saving, the Eco update rate, the On demand hotspot and Low or Medium transmit power; the Wi-Fi radio is by far the largest continuous power user, not the sensor.

Setting	Production value	Effect
Power saving	on	ESP32 CPU at 80 MHz instead of 240 MHz; LED strip refreshed only when a pixel changes; Wi-Fi modem sleep while the hotspot is off; short idle pauses in the main loop; quieter Serial output
Update rate	Eco	sensor every 5 s, open dashboard refreshes every 10 s
Hotspot	On demand	off until the BOOT / FLASH button is pressed; off again after 10 minutes without devices; radio fully off if there is no home Wi-Fi either
Transmit power	Low (Medium if the router is in another room)	less heat and interference

12.1 Hotspot modes

Mode	Behaviour
Automatic	on while there is no home Wi-Fi or it cannot be reached; off 30 s after home Wi-Fi connects
On demand	off until BOOT / FLASH is pressed; off after 10 min without devices; still on at start-up when no home Wi-Fi is saved
Always on	always on; most power and heat

12.2 Other heat sources

- **3.3 V regulator** next to the USB socket: it powers the controller and the sensor from 5 V, so warm is normal. Too hot to touch points to a short on 3V3 or the strip drawing power through the board.
- **Enclosure:** ventilation holes; not against the LED strip; out of direct sun.
- **LED strip:** brightness and LED count set the supply current (estimate in Display & sensor). Use a supply rated above that.

13. Firmware updates

Firmware can be updated over Wi-Fi from the dashboard (OTA) by an admin; settings, Wi-Fi, admin password, calibration, history and logs are kept.

13.1 Update over Wi-Fi (OTA)

1. In the Arduino IDE, with the same Tools settings as the installed build, choose **Sketch** → **Export Compiled Binary**.
2. In the sketch folder, open `build/<board>/` and take `water_level_indicator.ino.bin` (not the `.bootloader`, `.partitions` or `.merged` files).
3. Dashboard → Setup → **Firmware update** → Choose firmware file.
4. The dashboard checks the file before uploading: an ESP image, Water Tanks Monitor System firmware, built for this board (ESP32 or ESP8266), small enough for the free space. It shows the version and warns about a downgrade.
5. Confirm. A progress bar shows the upload; the device verifies the image, restarts, and the page confirms the new version.

The new firmware is written to the spare program slot. If the upload is interrupted or the image fails verification, the running firmware stays in place.

13.2 Size limits

Board and scheme	Program slot	OTA possible when
ESP32, Default 4MB with spiffs	1.2 MB	the sketch fits by USB
ESP32, Minimal SPIFFS	1.9 MB	the sketch fits by USB

Board and scheme	Program slot	OTA possible when
ESP8266 NodeMCU, 4MB (FS:2MB OTA:~1019KB)	about 1 MB shared by the running and the new program	the program is under about 500 KB

The Firmware update card shows the space available. If a build is too big, update once by USB.

13.3 Update by USB

Needed for the first installation, a change of partition scheme, or a build too big for OTA. Keep **Erase All Flash Before Sketch Upload: Disabled** (ESP32) or **Erase Flash: Only Sketch** (NodeMCU) so settings survive. Export the settings first as a precaution.

14. Backup and restore

Export the settings after setup and after every change; the file restores a device, or sets up an identical one, in a few seconds.

	Included	Not included
Export settings (JSON)	site names, usage and tank profile, calibration, capacity, low water alarm, analytics settings, units, LED count and brightness, strip options, trigger pulse, power, transmit power, hotspot mode, update rate, Settings ID, firmware version, home Wi-Fi name	Wi-Fi password, admin password, history, log

- **Export:** Setup → Admin & backup → Export settings. File name `wtms-settings-<building>-<tank>-<date>.json`.
- **Import:** Setup → Admin & backup → Import settings → choose the file → confirm. Wi-Fi is not changed. The Settings ID after import matches the file's ID if nothing else was changed.
- **History and log** are not in the backup. Download the log from the Log tab, and PDF reports from Analytics, to keep them.

15. Flash storage map

Settings live in a 512-byte EEPROM area and survive OTA updates and partition changes; history and log live in LittleFS and restart empty if its area is erased or moved.

15.1 EEPROM (512 bytes)

Offset	Size	Content
0	44 bytes	settings: calibration, alarms, display, analytics, power, units, update rate
64	about 104 bytes	home Wi-Fi name and password
200	8 bytes	boot counter
256	about 120 bytes	site: society, building, tank
384	24 bytes	tank profile: usage, location, shape, size, water depth
416	about 56 bytes	admin password: random salt and SHA-256 hash

Each block starts with its own marker. A block with a missing or wrong marker falls back to its defaults, so older firmware's settings are kept and new fields start at their defaults. The home Wi-Fi password is stored as entered, so anyone who can read the board's flash with a USB tool can read it.

15.2 LittleFS files

File	Record	Limit
/hist.bin, /hist.old.bin	8 bytes: time, distance (mm), filling flag	5,040 records each (7 days each, 14 days in total)
/fills.bin, /fills.old.bin	12 bytes: start, end, start and end distance	100 records each
/log.txt, /log.old.txt	one line per event	12 KB each

When a file reaches its limit it becomes the `.old` file and the older one is deleted, so the total stays at about 107 KB. Distances are stored rather than levels, so recalibrating also corrects old history.

16. Web API reference

The dashboard talks to the device through these HTTP endpoints on port 80; admin endpoints need the session token in an `auth` parameter (form field or query) and answer 403 without it.

Method	Path	Access	Parameters	Returns
GET	<code>/</code> , <code>/dashboard</code>	public		dashboard page
GET	<code>/api/status</code>	public	<code>auth</code> (optional), <code>live=1</code> (admin, fast readings)	JSON: level, distance, settings, Wi-Fi, site, profile, rate, auth state
GET	<code>/api/history</code>	public	<code>since</code> (unix time)	binary, 8-byte records
GET	<code>/api/fills</code>	public		binary, 12-byte records
POST	<code>/api/time</code>	public	<code>epoch</code>	sets the clock if unknown
GET	<code>/api/auth</code>	public		JSON: salt, one-time nonce
POST	<code>/api/login</code>	public	<code>proof</code>	JSON: token, mustChange
POST	<code>/api/logout</code>	session	<code>auth</code>	
POST	<code>/api/password</code>	admin	<code>hash</code> = SHA-256(salt + new password)	
POST	<code>/api/calibrate</code>	admin	<code>point=empty</code> or <code>full</code> , or <code>empty</code> and <code>full</code> (cm)	

Method	Path	Access	Parameters	Returns
POST	/api/settings	admin	any of: leds , brightness , lowAlarm , reversed , colorByLevel , trigUs , units , capacity , fillCm , nightStart , nightEnd , leakCm , perf , tx , apMode , rate , sensorMs , refreshS , defaults	
POST	/api/site	admin	org , building , tank	
POST	/api/profile	admin	usage , location , shape , preset , len , wid , hgt , dia , depth , gap (mm)	
GET	/api/scan	admin	start to begin a scan	JSON: networks
POST	/api/wifi/connect	admin	ssid , pass	
POST	/api/wifi/forget	admin		
POST	/api/restart	admin		
GET	/api/log	admin	download (optional)	text log
POST	/api/log/clear	admin		
POST	/api/history/clear	admin		
POST	/api/ota	admin	auth in the URL; multipart file	installs firmware and restarts

Any other address redirects to the dashboard, which makes phones and PCs open it automatically on the hotspot.

17. LED indications and log messages

The LED strip shows the device's state at a glance; the Log tab (admin) records the reasons behind it, and keeps them across restarts.

17.1 LED strip

What you see	Meaning
Red → green sweep at power-up	start-up test
Bar from the bottom, red to green	current water level
Bottom 3 LEDs blinking red	level below the low water alarm
Bottom LED blinking blue, rest off	sensor fault: no echo
Top LED blinking blue	hotspot is on
Whole strip flashes red 3 times	admin password reset with the BOOT / FLASH button

17.2 Log messages

Message	Meaning
<code>Boot #N ... restart cause: Power on</code>	power was cut or dipped (a weak supply shows many of these)
<code>restart cause: ... Watchdog, Exception, Crash, Brownout</code>	a fault restart (shown in red); check power and wiring
<code>Supply voltage (ESP8266)</code>	3.3 V rail at boot and every 5 minutes; under 3.0 V the supply is struggling
<code>Hotspot on / off (reason)</code>	when and why the hotspot changed
<code>Device joined / left hotspot</code>	phones and PCs on the hotspot, by MAC address
<code>Hotspot channel changed</code>	devices on the hotspot drop briefly (the ESP8266 hotspot follows the home Wi-Fi channel)

Message	Meaning
Home Wi-Fi lost / disconnect event	with the reason, e.g. signal lost or wrong password
Background retry stopped	a device joined the hotspot, so the home Wi-Fi retry paused
Health: ...	every 5 min: free memory, supply voltage, slowest loop, hotspot clients, signal
Main loop was blocked	the firmware stalled for more than 2 s
Sensor fault / Sensor OK	sensor stopped or started answering (at most once per 30 s)
Filling started / stopped	fill (motor) detected
Admin logged in / Admin login failed / password changed / reset	access events
Firmware update started / installed / failed	OTA events
Flash storage was blank and has been formatted	first start or after an erase; normal

Log times come from the internet or the browser; before the clock is known, entries show as `Boot #N +h:mm:ss`.

18. Troubleshooting

Start with the Log tab (admin) and the Serial Monitor at 115200 baud: most problems show there with a cause.

18.1 Sensor and level

Symptom	Likely cause	Fix
Bottom LED blinks blue, dashboard shows "no echo"	RX and TX swapped, loose wire, no common ground, or wrong pins	Check TRIG → sensor RX, ECHO → sensor TX, GND to GND; on ESP32 D15 / D2, on NodeMCU D5 / D6. Replace suspect jumper wires.
"No echo" with correct wiring	trigger pulse too short for the sensor, or sensor in serial mode	Set trigger pulse to 100 µs; check that position R19 on the sensor board is empty
"No echo" only when the tank is nearly full	water inside the blind zone	Raise the probe on a stand pipe so the full line is 25 cm or more below it
Level jumps or reads too high	echo from a pipe, wall, ladder or the stand pipe; probe not vertical	Move the probe to the centre, point it straight down, use a stand pipe at least 75 mm wide
Level always 0 % or 100 %, or wrong	not calibrated, or calibrated for another tank depth	Setup → Calibration (slider or measurements); check "Tank at a glance" warnings
Level changes slowly	Eco update rate (about 15 s)	Normal; use Balanced or Responsive if faster response is needed
Litres wrong but % right	capacity or water depth wrong; tapered tank	Enter the rated capacity and measured water depth; litres assume straight walls

18.2 LED strip

Symptom	Likely cause	Fix
No LEDs, or random colours	no common ground, missing 330 Ω, data on wrong pin	Join grounds; ESP32 G16, NodeMCU D2; add the resistor near the strip

Symptom	Likely cause	Fix
Only part of the strip lights at full level	LED count setting too low	Display & sensor → Number of LEDs
Bar fills from the wrong end	strip mounted top-down	Turn on "Strip mounted top-down"
Far end dim or yellow	voltage drop along the strip	Feed 5 V and GND to both ends
Board resets when LEDs are bright	supply too weak	Use a larger 5 V supply or lower the brightness

18.3 Wi-Fi, hotspot and dashboard

Symptom	Likely cause	Fix
Hotspot does not appear	On demand mode	Press BOOT (ESP32) or FLASH (NodeMCU); the top LED blinks blue when it is on
Hotspot disappears	connected to home Wi-Fi (by design), or the board restarts	Check the log: "Hotspot off (connected to home Wi-Fi)" is normal; repeated "Boot #" means power problems
Connected to the hotspot, page does not load	https address or mobile data in use	Open http://192.168.4.1 exactly; turn off mobile data
<code>.local</code> address does not open	some Android phones do not support <code>.local</code>	Use the IP address from Wi-Fi settings
Dashboard unreachable after a router change	saved network no longer exists	Wait about 20 s (automatic mode) or press BOOT, join the hotspot, choose the new network
"Admin session ended"	device restarted or 12 h idle	Log in again
Forgotten admin password		Hold BOOT / FLASH 10 s until the strip flashes red; log in with <code>admin</code>
"Too many attempts"	5 wrong passwords	Wait 1 minute

18.4 Flashing, storage and updates

Symptom	Likely cause	Fix
Settings, Wi-Fi or calibration lost after a USB upload	Erase All Flash enabled	Set it to Disabled (ESP32) or Erase Flash: Only Sketch (NodeMCU); import the settings backup

Symptom	Likely cause	Fix
Serial shows <code>esp_littlefs ...</code> Corrupted dir pair (firmware 2.8.0 and older)	blank storage area	Normal after an erase; formatted automatically
Log storage: Memory only	partition scheme without a spiffs area	Choose Default 4MB with spiffs or Minimal SPIFFS (ESP32), or Flash Size with FS (NodeMCU)
"Sketch too big"	program larger than the slot	ESP32: Minimal SPIFFS (1.9MB APP with OTA)
ESP32 upload fails with "wrong boot mode"	sensor holding D2 high	Unplug the sensor connector, upload, plug it back; check D15 / D2 are not swapped
OTA refused: wrong board or not firmware	wrong <code>.bin</code> file	Use <code>water_level_indicator.ino.bin</code> exported for this board
OTA refused: not enough space	program too big for OTA (mainly NodeMCU)	Update by USB
OTA upload interrupted	Wi-Fi drop	Nothing changed; try again closer to the router

18.5 Analytics, reports and heat

Symptom	Likely cause	Fix
Analytics empty or "clock unknown"	no internet time yet	Open the dashboard once (the browser sets the clock) or connect to home Wi-Fi
No fills detected	fill slower than the threshold	Lower the fill detection threshold
False fills	ripples or noise	Raise the threshold; check probe position

Symptom	Likely cause	Fix
"Possible leak" at night	slow steady drop: a leak, a dripping tap or float valve, or evaporation	Check taps, float valve and pipes; adjust the night window or limit if the site uses water at night
PDF does not download	hotspot pop-up window blocks downloads	Open http://192.168.4.1 in the normal browser
Board runs hot	Wi-Fi at full power, performance mode, hotspot always on	Power saving on, Eco rate, hotspot On demand, transmit power Low

19. Maintenance checklist

A short check once a month keeps the readings trustworthy; the log and the Tank at a glance card show most problems before anyone notices them on the strip.

Monthly

- ☐ Log tab: no new fault restarts (Watchdog, Exception, Brownout) and no repeated "Power on" boots.
- ☐ Tank at a glance: no warnings; level matches a quick visual check of the tank.
- ☐ Analytics: fills and night checks look plausible; investigate any "Possible leak".
- ☐ Download a PDF report for the records.

Every 6 months

- ☐ Clean the probe face (dust, cobwebs, condensation) with a dry cloth.
- ☐ Check the probe is still vertical and clear of pipes; check cables and terminals for corrosion.
- ☐ Check the enclosure for water, insects and heat marks; ventilation holes clear.
- ☐ Verify calibration against a measured level; recalibrate with the slider if it is off by more than 1 to 2 cm.
- ☐ Install any new firmware (OTA) and export the settings afterwards.

After any change (tank, sensor position, router, firmware): recalibrate if needed and export the settings.

20. Licence, copyright and support

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- **Website:** woodyouloveit.com
- **Source code and issues:** github.com/ChanchalSakardeQH/WATERTANK-LEVEL-INDICATOR-with-Dashboard

When asking for support, include:

1. Board (ESP32 or NodeMCU) and firmware version (Setup → Device).
2. The downloaded log (Log tab → Download).
3. The exported settings file, or its Settings ID.
4. A photo of the wiring and of the probe mounting, if the problem is with readings.